EE 301 – LAB 4

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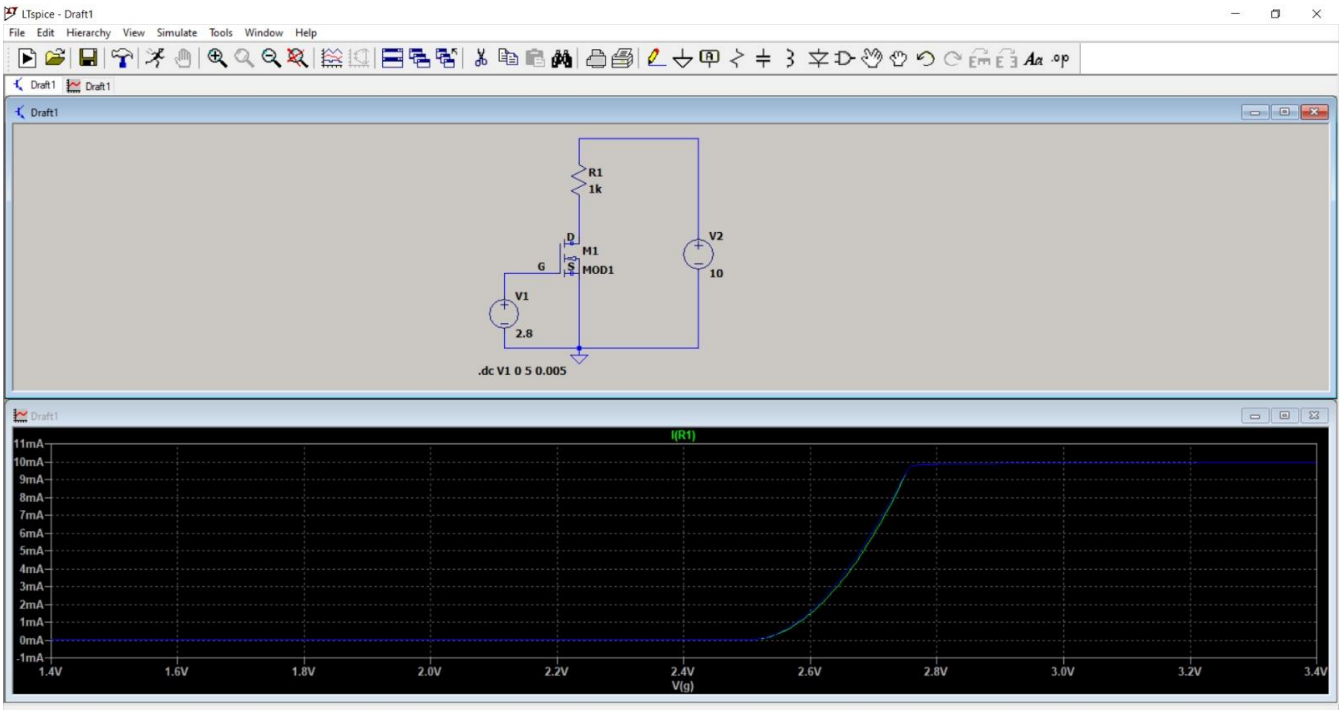
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Introduction

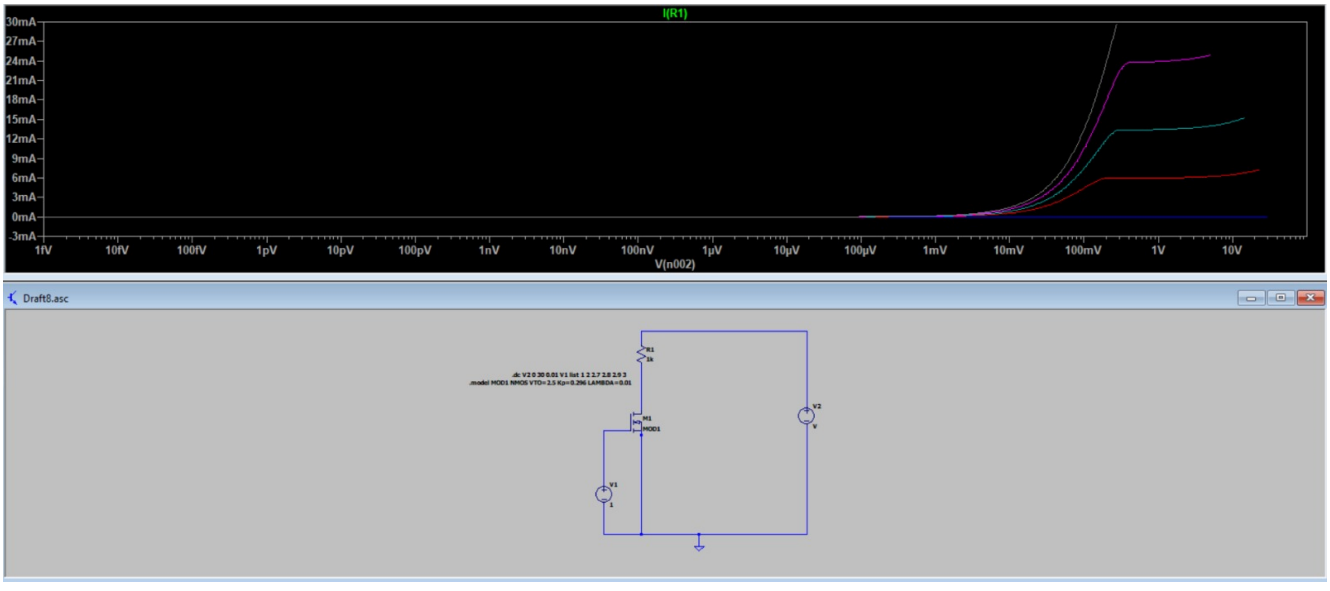
For that lab we are diving into MOS transistors. We'll focus on two types: NMOS (2N7000) and PMOS (ZVP3306A). Through hands-on experiments, we're measuring current gain, checking how transistors behave in different situations, and getting practical with these electronic building blocks. This hands-on experience sets the stage for understanding how these components work in electronic circuits.

Preliminary Work

1- For the preliminary work, we need to construct the circuit 2.1 in LTSpice and plot current through the R1.



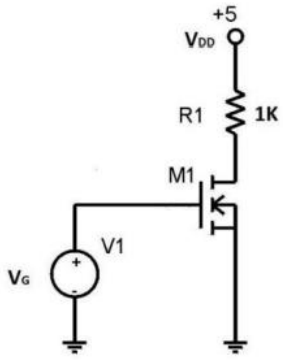
2- For the next step we set LAMBDA to 0.01. Sweep VDD from 0 to 30V and plot drain current as a function of drain-source voltage for VG= 1, 2, 2.7, 2.8, 2.9 and 3 V.



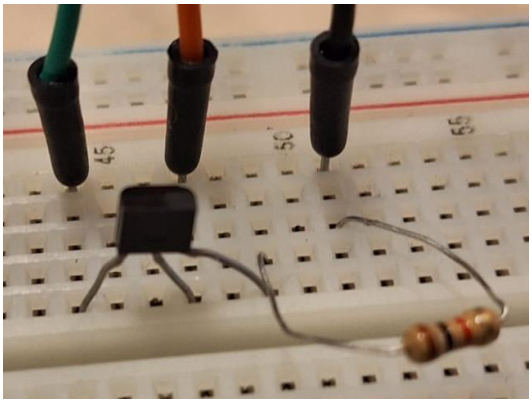
Lab Tasks

Task 1.

1- First, we need to construct circuit 3.1 with 2N7000 NMOS transistor, fixed +5V on V_{DD} and 1KΩ on the R1.



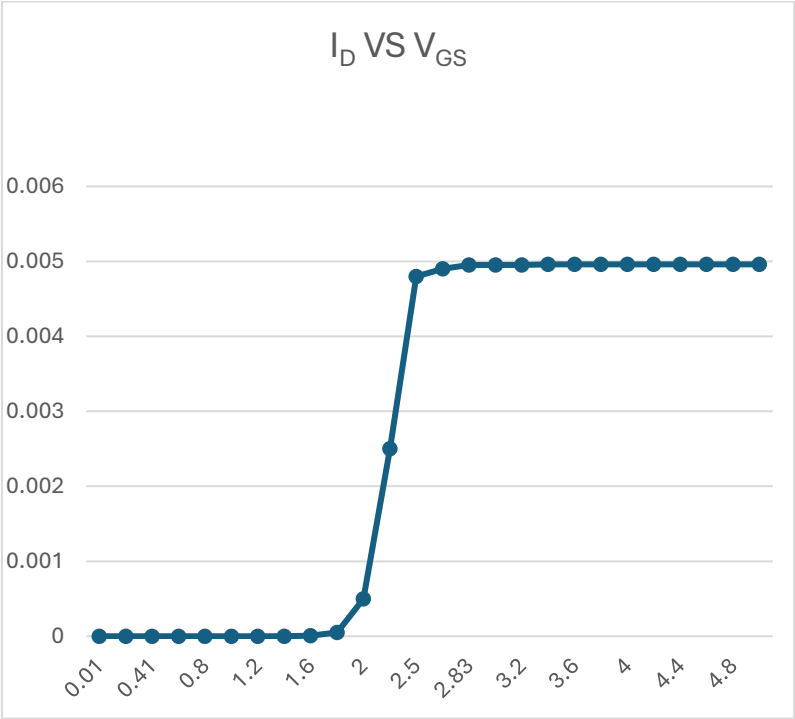
Cir 3.1



2- We have constructed circuit 3.1 as shown on the figure X and connect V_G to the variable voltage source and vary that V_G from 0 to 5V with 0.2V steps, and measure voltage drop over drain resistor to plot I_D vs V_{GS}.

3- Here is the output of the measurements below.

V_G	R (V) 1k	V_{GS}	V_{DG}	I_D
0	0	0.01	5	0
0.2	0	0.195	4.88	0
0.4	0	0.41	4.6	0
0.6	0	0.56	4.5	0
0.8	0	0.8	4.2	0
1	0	1	4	0
1.2	0	1.2	3.8	0
1.4	0	1.4	3.6	0
1.6	0.008	1.6	3.4	0.000008
1.8	0.05	1.8	3.2	0.00005
2	0.5	2	2.5	0.0005
2.2	2.5	2.2	-1	0.0025
2.4	4.8	2.5	-2.4	0.0048
2.6	4.9	2.6	-2.6	0.0049
2.8	4.95	2.83	-2.8	0.00495
3	4.95	3	-3	0.00495
3.2	4.95	3.2	-3.2	0.00495
3.4	4.96	3.43	-3.45	0.00496
3.6	4.96	3.6	-3.6	0.00496
3.8	4.96	3.8	-3.8	0.00496
4	4.96	4	-4	0.00496
4.2	4.96	4.2	-4.2	0.00496
4.4	4.96	4.4	-4.4	0.00496
4.6	4.96	4.6	-4.6	0.00496
4.8	4.96	4.8	-4.8	0.00496
5	4.96	5	-5	0.00496



4- To get 1mA on the drain current, we needed to set V_G to 1.8V. Then we measure drain current before and after heating the transistor. Then we have measured,

I_D, Before heating up the transistor: 1.27mA

I_D, After heating up the transistor: 1.89mA

5- Now we set V_G to 4V and supply voltage (V_{DD}) to 3V. Measure I_D and V_{DS}. Then determine what is the operation regime of the transistor.

For the V_G at 4V and V_{DD} at 3V we have got

V_{DS} = 0.007V and I_D = 0.007V at Saturation.

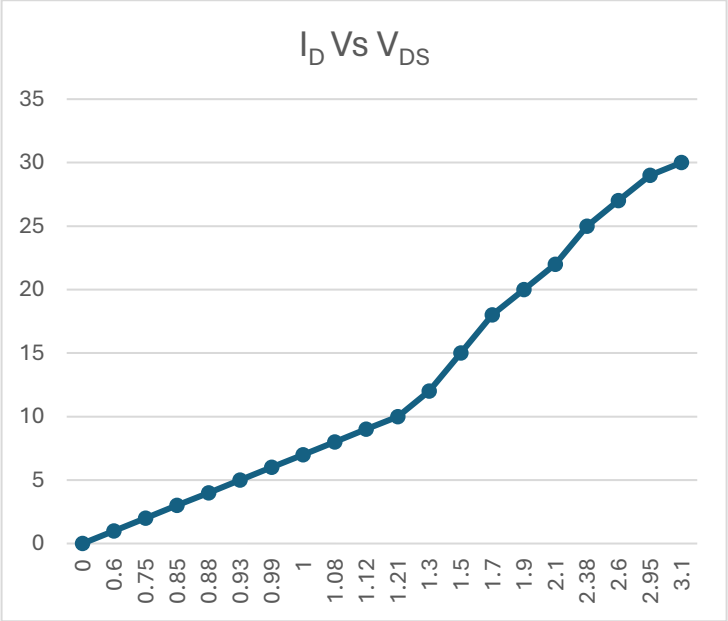
For the V_G at -3V and V_{DD} at 5V we have got

V_{DS} = 0V and I_D = 0.007V at Cut-off.

Task 2

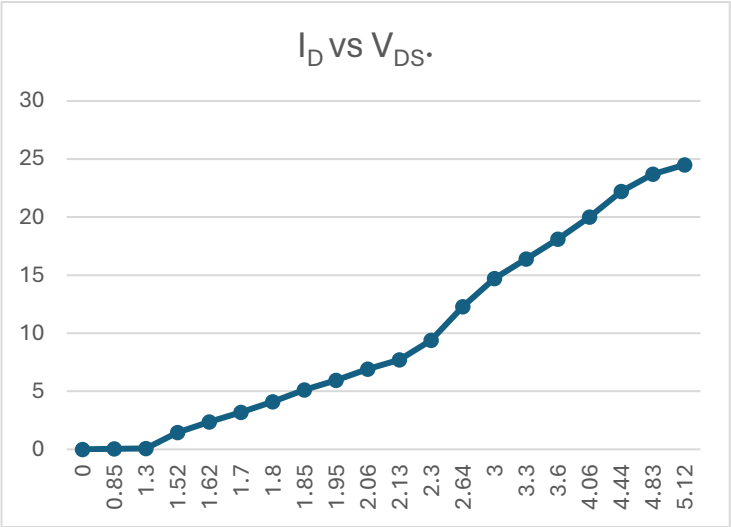
- We are continuing on the same circuit (3.1) and connect V_{DD} to a variable source such that you can change it from 0 to 30V.
- Set V_G such that I_D is 0.5mA, which is 2V for V_G and plotted I_D vs V_{DS}.

V_G	V_D	V_{DS}	R (V)	I_D (mA)
2	0	0	0	0
2	1	0.4	0.6	0.6
2	2	1	0.75	0.75
2	3	2.15	0.85	0.85
2	4	3.13	0.88	0.88
2	5	4	0.93	0.93
2	6	5	0.99	0.99
2	7	5.95	1	1
2	8	6.8	1.08	1.08
2	9	7.5	1.12	1.12
2	10	9	1.21	1.21
2	12	10.5	1.3	1.3
2	15	13.4	1.5	1.5
2	18	16	1.7	1.7
2	20	18	1.9	1.9
2	22	19.66	2.1	2.1
2	25	22.3	2.38	2.38
2	27	24	2.6	2.6
2	29	25.75	2.95	2.95
2	30	26.5	3.1	3.1



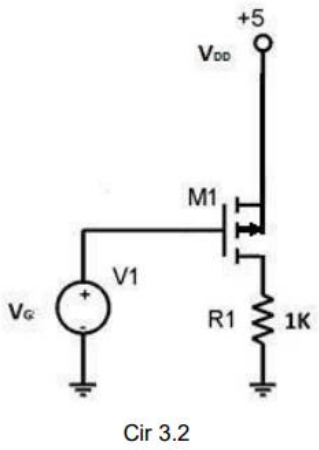
-Then we set V_G such that I_D is 2mA, which is 2.1V for V_G and plotted I_D vs V_{DS}.

V_G	V_D	V_{DS}	R (V)	I_D (mA)
2.1	0	0	0	0
2.1	1	0.04	0.85	0.85
2.1	2	0.06	1.3	1.3
2.1	3	1.45	1.52	1.52
2.1	4	2.37	1.62	1.62
2.1	5	3.2	1.7	1.7
2.1	6	4.1	1.8	1.8
2.1	7	5.13	1.85	1.85
2.1	8	5.95	1.95	1.95
2.1	9	6.9	2.06	2.06
2.1	10	7.7	2.13	2.13
2.1	12	9.4	2.3	2.3
2.1	15	12.3	2.64	2.64
2.1	18	14.7	3	3
2.1	20	16.4	3.3	3.3
2.1	22	18.1	3.6	3.6
2.1	25	20	4.06	4.06
2.1	27	22.2	4.44	4.44
2.1	29	23.7	4.83	4.83
2.1	30	24.5	5.12	5.12

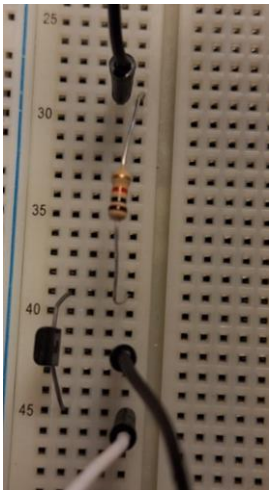


Task 3

- We have constructed the circuit shown in Cir 3.2 with ZVP3306A transistor with fixed V_{DD} +5V output of the power supply and connected V_G to the variable voltage source.

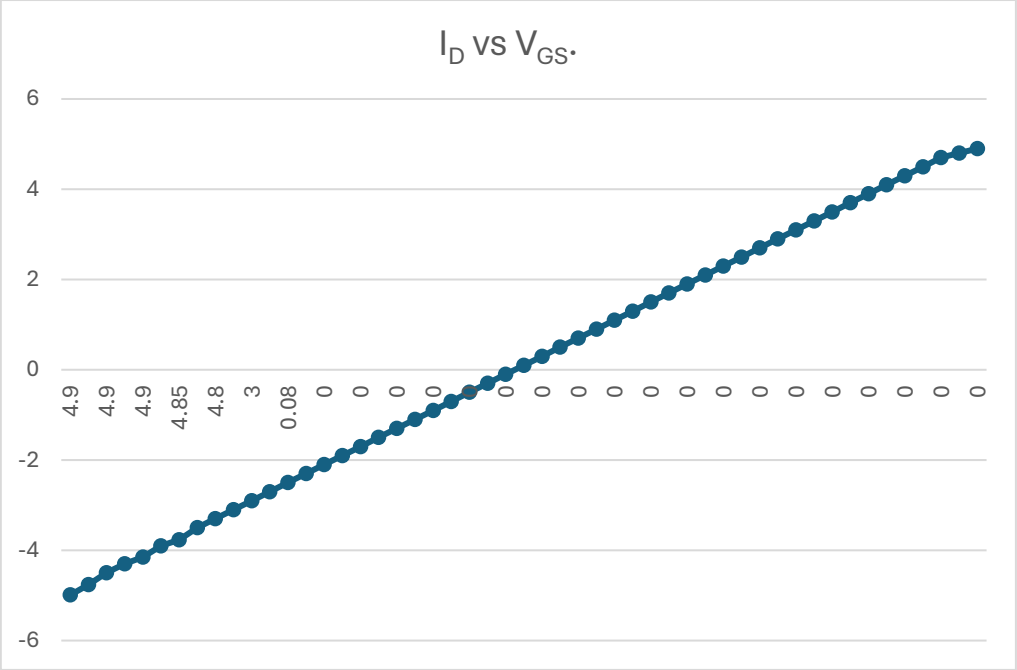


Cir 3.2



- Then varied V_G from 0 to 10V with 0.2V steps and measured voltage drops over drain resistor. Then Plotted I_D vs V_{GS}.

VD	VG	R (V)	VGS
5	0	4.9	-4.99
5	0.2	4.9	-4.76
5	0.4	4.9	-4.5
5	0.6	4.9	-4.3
5	0.8	4.9	-4.15
5	1	4.9	-3.9
5	1.2	4.85	-3.77
5	1.4	4.8	-3.5
5	1.6	4.8	-3.3
5	1.8	4.76	-3.1
5	2	3	-2.9
5	2.2	0.8	-2.7
5	2.4	0.08	-2.5
5	2.6	0.013	-2.3
5	2.8	0	-2.1
5	3	0	-1.9
5	3.2	0	-1.7
5	3.4	0	-1.5
5	3.6	0	-1.3
5	3.8	0	-1.1
5	4	0	-0.9
5	4.2	0	-0.7
5	4.4	0	-0.5
5	4.6	0	-0.3
5	4.8	0	-0.1
5	5	0	0.1
5	5.2	0	0.3
5	5.4	0	0.5
5	5.6	0	0.7
5	5.8	0	0.9
5	6	0	1.1
5	6.2	0	1.3
5	6.4	0	1.5
5	6.6	0	1.7
5	6.8	0	1.9
5	7	0	2.1
5	7.2	0	2.3
5	7.4	0	2.5
5	7.6	0	2.7
5	7.8	0	2.9
5	8	0	3.1
5	8.2	0	3.3
5	8.4	0	3.5
5	8.6	0	3.7
5	8.8	0	3.9
5	9	0	4.1
5	9.2	0	4.3
5	9.4	0	4.5
5	9.6	0	4.7
5	9.8	0	4.8
5	10	0	4.9



Conclusion

To sum up, our experiments with MOS transistors, we've gained valuable insights into their behaviors and characteristics. Working hands-on with NMOS (2N7000) and PMOS (ZVP3306A) transistors, we measured current gain, observed responses under varying conditions, and gained hands-on experience for our practical skills. This experiment provides an understanding of semiconductor devices and their role in electronic circuits.